PEPPER EQUIPMENT CORPORATION / MONTE VISTA, COLORADO

KUKA Industrial Automation

Agricultural AI-robotics R&D, KRL interface code, and cell integration



KUKA Delta robot cell handling potatoes over a roller conveyor. Workplace photo, November 2022.

Mechatronics Implementation Engineer | Full-time

September 19, 2022 - March 1, 2023. Contributed to an agricultural AI-robotics R&D project, programming and testing the interface code between the AI system and KUKA Delta robot.

My contribution

AI-to-robot interface programming and testing; robot/conveyor coordination; automatic external controls; installation and testing of pneumatic and mechanical systems associated with the robot cell.

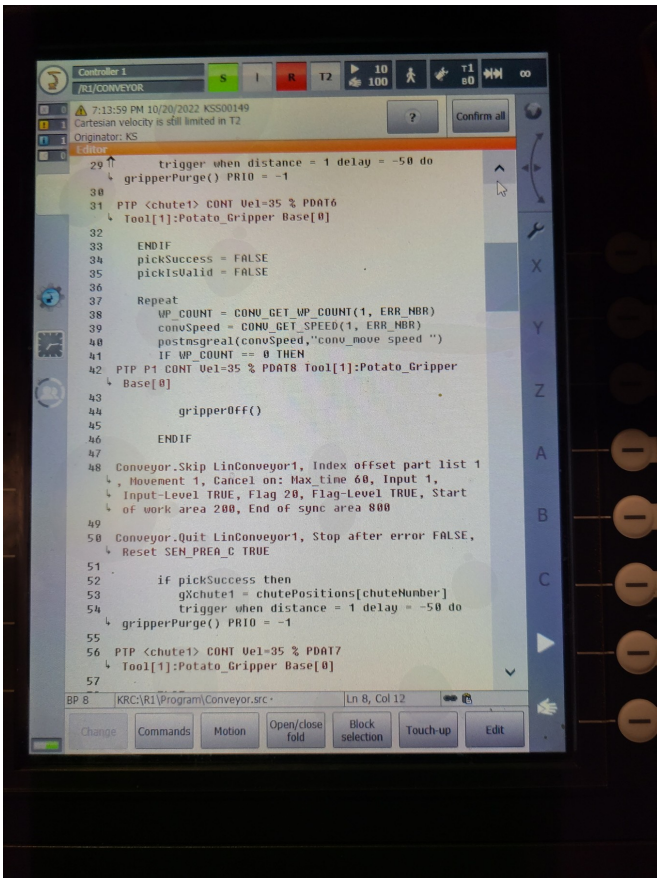
Industrial robotics | KUKA programming | I/O integration | Pneumatics

Dakshitha (Dak) Amarasinghe

dakshithaama1994@gmail.com | LinkedIn profile

01 / APPLICATION PROGRAMMING

Robot and conveyor coordination





Left: pendant view of Conveyor.src, showing motion, conveyor, and gripper logic. Right: physical robot and product-handling area.

AI-to-robot interface and process sequencing

Programmed and tested interface code between the AI system and KUKA Delta robot, as confirmed in the company experience statement. Robot and conveyor work connected motion targets, gripper actuation, conveyor state, and placement behavior.

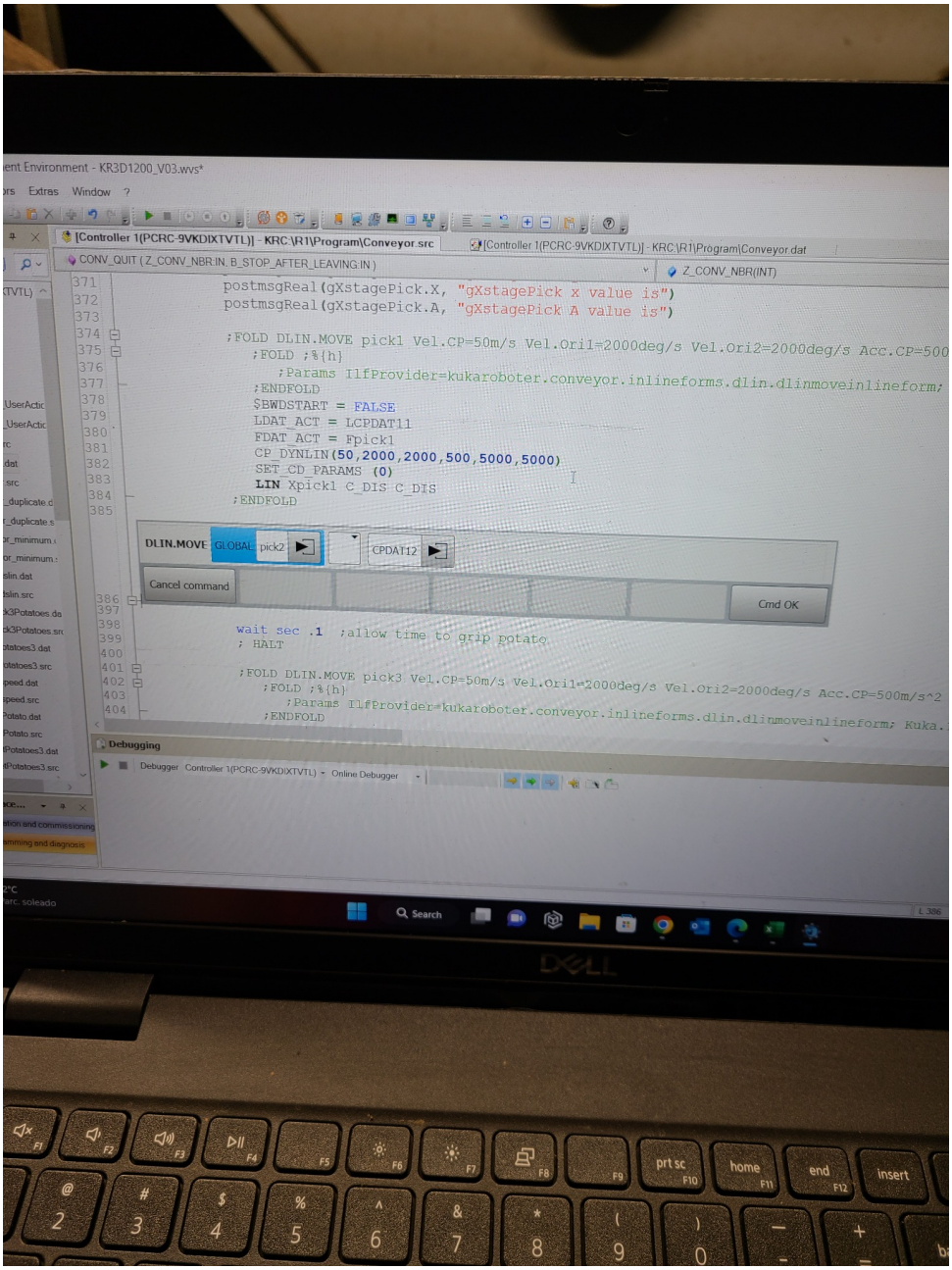
What the program photograph shows

Visible statements include conveyor workpiece-count and speed queries, conditional pick-success handling, chute-position selection, PTP motions with CONT, and gripper functions.

Evidence boundary: supplied program photographs show excerpts, not a complete controller backup. No cycle-time or throughput improvement percentage is claimed.

01 / APPLICATION PROGRAMMING

Motion programming detail

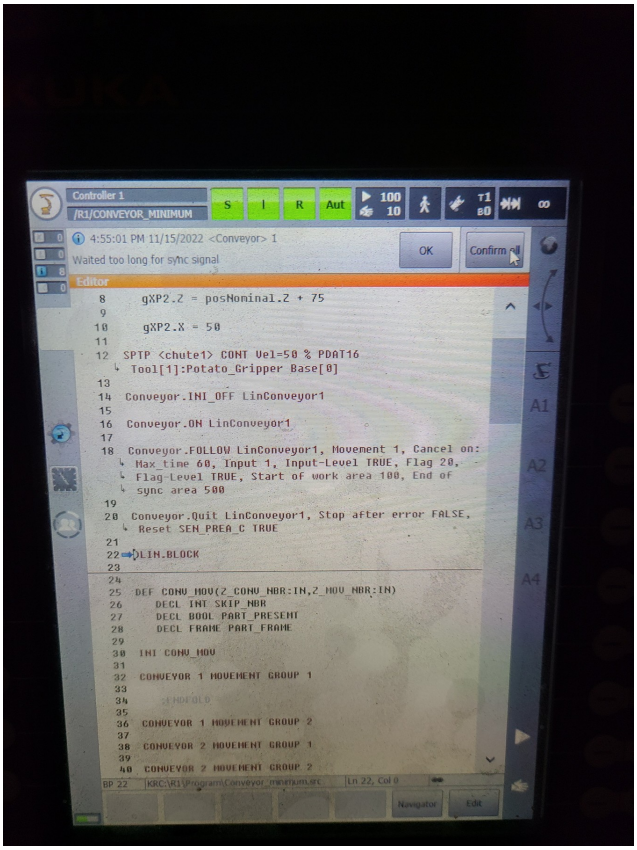


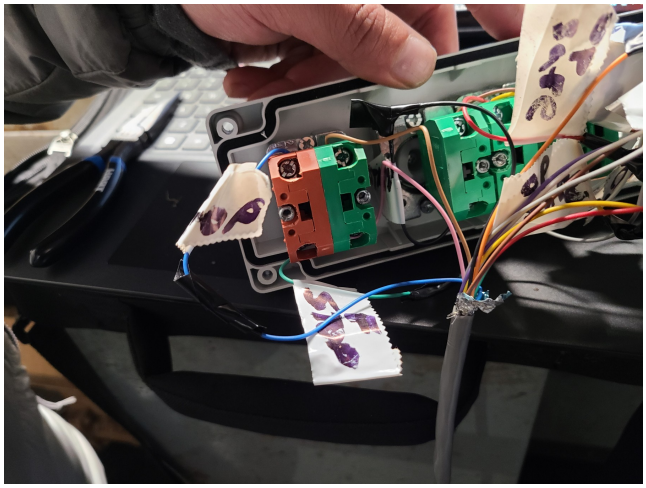
Additional program photograph: Conveyor.src in the development environment, showing DLIN.MOVE inline forms, LIN motion, motion-data assignments, and a wait instruction annotated for potato gripping.

This view complements the pendant and conveyor photos on the preceding page. Displayed parameter values are code context, not independently measured speed or performance results.

02 / WIRING & CONTROLLER I/O

Automatic external operation





Controller program view and a wired switch assembly photographed during implementation. Switch functions are not assigned from photographs alone.

Operator interface integration

Wired, configured, and programmed automatic external operation using four switches, together with an emergency-stop switch and an operator-safety switch. This work spanned physical connections and robot-controller configuration.

Electrical and software integration

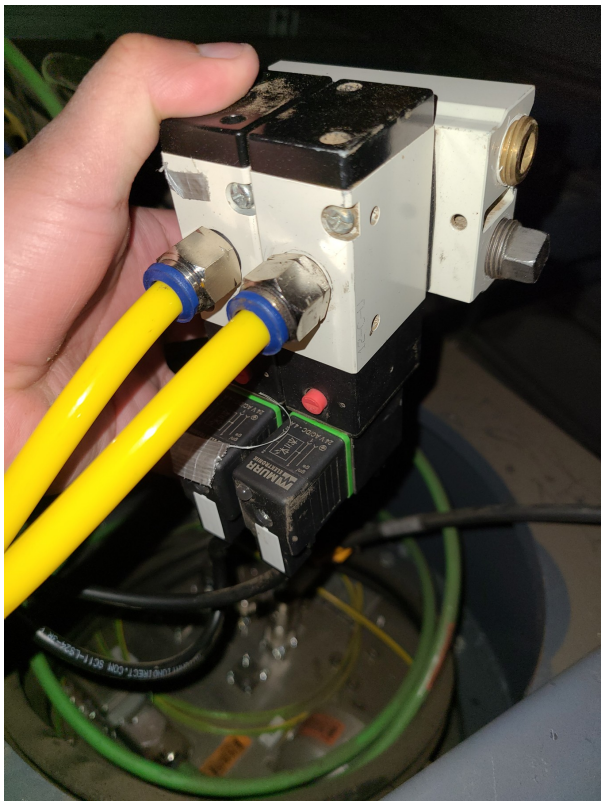
The supplied image set includes controller I/O screens, connector wiring, and configuration views. The integration task required relating external device signals to the robot control interface while supporting intended operating sequences.

Scope of the documented work

The portfolio describes implementation and testing as stated in the employment record. It does not establish a certified safety architecture, validated safety performance level, or specific switch-to-signal mapping.

03 / ACTUATION & HARDWARE

Electro-pneumatic integration





Pneumatic valve/solenoid assembly with tubing, and a digital pneumatic instrument in the cell. Component settings are not inferred from these photos.

Electro-pneumatic installation and testing

Installed and tested electro-pneumatic controls for the robot cell, combining electrical command connections with pneumatic hardware and tubing. The program photographs also show gripper activation and purge functions within the handling sequence.

A multidisciplinary integration task

The robot program, electrical I/O, and pneumatic response form one operational chain. This project required working across those interfaces while configuring and testing the cell for the potato-handling application.

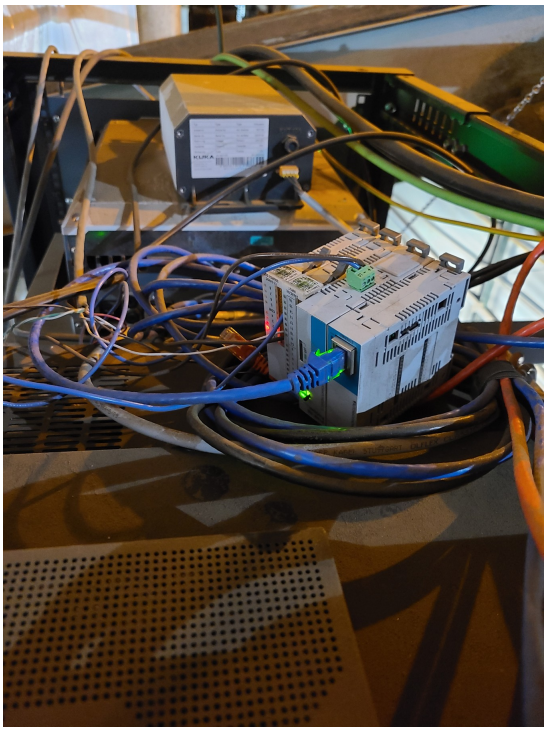


Hands-on component integration

Additional workplace photograph of a pneumatic component and attached electrical wiring. This complements the valve, tubing, and controller evidence.

04 / SYSTEMS PERSPECTIVE

Robot-cell systems integration





Cell wiring and pneumatic integration photographed during implementation. These are workplace evidence images rather than as-built circuit schematics.

Robot-cell mechanical integration

Installed and tested mechanical systems associated with the robot cell, as confirmed by the company experience statement. This responsibility is distinct from additional welding, grinding, and drilling assignments performed elsewhere for the company.

Conceptual functional relationships

Product conveyor -> robot handling area -> placement destination. Robot program -> controller I/O -> pneumatic gripper. External operator controls -> robot operating interface. This summary explains system roles; it is not an electrical or safety schematic.

Engineering value

The contribution spans interface software, physical wiring, pneumatic actuation, and robot-cell mechanical integration. It demonstrates the ability to work across subsystem boundaries during industrial R&D.

05 / OUTCOMES & CAPABILITIES

Industrial automation experience

Work performed during R&D

AI-to-KUKA interface programming and testing; robot/conveyor programming; external control wiring and configuration; pneumatic and robot-cell mechanical installation and testing.

Skills demonstrated

KRL interface programming, motion-sequence coordination, controller I/O configuration, electrical wiring, pneumatic integration, and mechanical-system testing. The project shows applied mechatronics experience in agricultural automation R&D.

SUPPLIED ROBOT DATASHEET

Model documented	KUKA KR 3 D1200 HM
Axes / mounting	4 controlled axes / ceiling mounting
Working diameter	1200 mm
Rated payload	3 kg
Pose repeatability	+/- 0.1 mm, manufacturer specification

Specifications above come from the supplied datasheet (V6.1, 27 May 2022), not measurements of this workcell. The datasheet does not by itself verify the installed robot/controller configuration. Its conditional 6 kg maximum payload is not presented as the normal operating payload.

Evidence and attribution

Based on the company work-experience statement supplied by the author, the resume description, 38 workplace photographs dated October 2022-February 2023, and the supplied KUKA datasheet. Company-confirmed responsibilities include AI/robot interface code and pneumatic/mechanical installation and testing.

This was an R&D project discontinued before prototype completion; it is not presented as a completed production deployment. No numerical production gains or safety certification are claimed.

06 / EMPLOYER-CONFIRMED EXPERIENCE

Training, scope, and project context

KUKA training and KRL certification

Received KUKA Robotics Language (KRL) training from KUKA Robotics Corporation and earned KRL certification, as stated in the company work-experience paragraph supplied by the author.

Responsibilities confirmed by the company

Responsible for programming and testing interface code between the AI system and KUKA Delta robot, and for installation and testing of pneumatic and mechanical systems associated with the robot cell.

Project status

The agricultural AI-robotics project was closed during R&D before prototype completion. According to the company statement, the estimated time and resources required exceeded the initial expectations and stakeholder willingness to continue. The portfolio presents the work performed during development, not a completed production system.

Additional company work - separate from this cell

Performed mechanical-part manufacturing using MIG welding, grinding, drilling, and other shop equipment on additional company assignments. These activities were not part of the robot-workcell procedure.

Professional recognition

The company statement describes performance as satisfactory to management and expresses appreciation for initiative, cooperation, effort, and commitment to the project.

EMPLOYMENT RECORD

Mechatronics Implementation Engineer
Pepper Equipment Corporation, Monte Vista, Colorado
September 19, 2022 - March 1, 2023

Attribution: employer statements above are paraphrased from the work-experience paragraph provided by the author. Training certificates and the signed company letter are not reproduced in this portfolio.